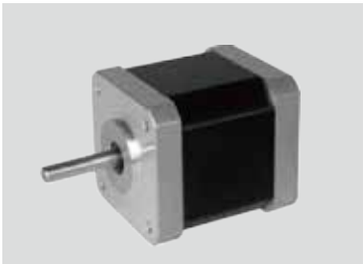
# PHB42S Series

## Key Features

- High accuracy
- High Torque
- Low Inertia

## Option

Linear, Encoder, IP 65, Hollow, Gearbox, Brake



| | |
|---|---|
| Step Angle | 1.8° |
| Step Angle Accuracy | ±5% (full step, no load) |
| Resistance Accuracy | ±10% |
| Inductance Accuracy | ±20% |
| Temperature Rise | 80 °C Max.(rated current, 2 phase on) |
| Ambient Temperature | -20 °C ~ +50 °C |
| Insulation Resistance | 100 MΩ Min., 500VDC |
| Dielectric Strength | 500 VAC for one minute |
| Shaft Radial Play | 0.02 Max. (450 g-load) |
| Shaft Axial Play | 0.08 Max. (450 g-load) |

## Specifications

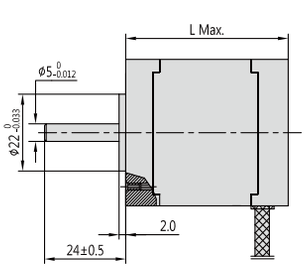
- Bi-polar

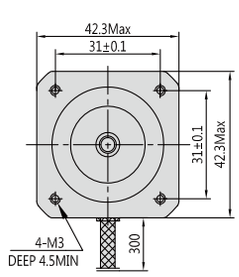
| Series & Length | Model Number | Holding Torque | | Rated Current | Resistance per Phase | Inductance per Phase | Detent Torque | | Rotor Inertia | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | N.cm | oz-in | A | ohm | mH | N.cm | oz-in | g.cm² | oz-in² |
| PHB42S20 20mm | PHB42S20-410 | 10 | 14 | 0.8 | 3.8 | 4.5 | 1.0 | 1.4 | 24 | 0.132 |
| PHB42S28 28mm | PHB42S28-406 | 12 | 16.8 | 0.6 | 8.0 | 10 | 1.2 | 1.68 | 30 | 0.165 |
| PHB42S34 34mm | PHB42S34-401 | 28 | 39.2 | 1.3 | 2.4 | 2.8 | 1.6 | 2.24 | 34 | 0.187 |
| | PHB42S34-402 | 28 | 39.2 | 1.7 | 1.2 | 1.8 | 1.6 | 2.24 | 34 | 0.187 |
| PHB42S40 40mm | PHB42S40-401 | 40 | 56 | 1.3 | 2.5 | 5.0 | 2.2 | 3.08 | 54 | 0.297 |
| | PHB42S40-402 | 40 | 56 | 1.7 | 1.5 | 2.8 | 2.2 | 3.08 | 54 | 0.297 |
| PHB42S48 48mm | PHB42S48-401 | 52 | 72.8 | 1.3 | 3.2 | 5.5 | 2.6 | 3.64 | 68 | 0.374 |
| | PHB42S48-402 | 52 | 72.8 | 1.7 | 1.8 | 3.2 | 2.6 | 3.64 | 68 | 0.374 |
| | PHB42S48-403 | 52 | 72.8 | 2.3 | 1.2 | 1.6 | 2.6 | 3.64 | 68 | 0.374 |
| PHB42S60 60mm | PHB42S60-403 | 70 | 98 | 2.3 | 1.4 | 2.5 | 3.6 | 5.04 | 80 | 0.440 |
| PHB42S63 63mm | PHB42S63-403 | 80 | 112 | 2.3 | 1.5 | 2.2 | 4.0 | 5.6 | 85 | 0.468 |

- Uni-polar

| Series & Length | Model Number | Holding Torque | | Rated Current | Resistance per Phase | Inductance per Phase | Detent Torque | | Rotor Inertia | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | N.cm | oz-in | A | ohm | mH | N.cm | oz-in | g.cm² | oz-in² |
| PHB42S34 34mm | PHB42S34-630 | 21 | 29.4 | 0.4 | 30 | 18 | 1.6 | 2.24 | 34 | 0.187 |
| PHB42S40 40mm | PHB42S40-630 | 28 | 39.2 | 0.4 | 30 | 28 | 2.2 | 3.08 | 54 | 0.297 |
| PHB42S48 48mm | PHB42S48-630 | 34 | 47.6 | 0.4 | 30 | 38 | 2.6 | 3.64 | 68 | 0.374 |

## Mechanical Dimension




| Series | L | Mass |
|---|---|---|
| | mm (in.) | g (lb.) |
| PHB42S20 | 20 (0.79) | 120 (0.26) |
| PHB42S28 | 28 (1.10) | 150 (0.33) |
| PHB42S34 | 34 (1.34) | 220 (0.49) |
| PHB42S40 | 40 (1.57) | 280 (0.62) |
| PHB42S48 | 48 (1.89) | 350 (0.77) |
| PHB42S60 | 60 (2.36) | 480 (1.06) |
| PHB42S63 | 63 (2.48) | 500 (1.10) |